1

* **Tokenization** :->

1) word tokenizer -> General/default tokenizer in NLP
handles punctuation intelligently. (?, !, .)
2) WhitespaceTokenizer -> splits based on spaces
3) TreebankWordTokenizer -> Linguistically advanced tokenizer ->
ex :-> don't
"do", "n't"
4) Tweet Tokenizer :-> Designed for Social Media Text. where language is messy.
" I am @[redacted] and #new 😊 "
@, #, images 😊
5) **MWETokenizer** (multi-word expression Tokenizer)
combines multiple words (phrases) into single token.

New york

"New_York"

* **Tokenization** :-> splitting the text into smaller units called tokens, so that the machine can process it
* **Stemming** :-> Stemming reduces words to their root/base form.

(PorterStemmer)

ex:- running -> run
playing -> play
studies -> studi

* Lemmatization :->
Lemmatization reduces a word to its dictionary form (lemma) using (meaning + grammar)

WordNetLemmatizer

playing -> play
better -> good
studies -> study.

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Stemmer

Porter Stemmer → oldest version (Basic)
Snowball Stemmer → advanced version

Working → uses a set of fixed rules
- removes suffixes like ing, ed, ly, ies etc.

*) Less Accurate

WordNetLemmatizer → it is a lexical database WordNet, which contains relationships between words (synonyms, meanings, grammar etc).

2

* Text vectorization → Converting text into numerical form.

*1) Bag of Words → counts how many times each word appears in vocabulary. (simple and easy)

* 2) TF-IDF → Improves bag of words by giving importance to meaningful words.

Unique → Valuable
freq → Less val.

TF → How often word appears in a document $TF = (\text{No. WF} / \text{Total No. of terms in doc})$

IDF → How rare/unique word across all DOCS.

TF-IDF = TF × IDF

$$IDF = \frac{(\text{Total No. of DOCS})}{(\text{No. of DOC containing the word})}$$

3) Word2Vec → neural network based method that captures the meaning of words / converting text into vectors or in numerical form.

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3

* Text cleaning :→ Text cleaning means removing unnecessary parts of texts.
  Includes :→ Lowercasing
  Punctuations
  (news2563) → news — remove numbers
  remove extra spaces
  (#, @, $) — remove special symbols
* Label encoding :→ converting categorical columns into numbers.
* Stop-word Removal :→ preprocessing technique where very common words are removed from text due to these words not add any important meanings.
  is, the, and
  of, to, in, a, an, for

4

* transformer :→ deep learning networks that uses for processes the sequential data like → text. It can process data parallel (parallel processing).
* Positional embeddings :→
  tokenization to → token ids → strips the order of words → to resolve this add information about abs/Relative position of tokens back into the sentence embeddings.
* Attention mechanism :→ It focuses on relevant parts of the input sequence for each output.
  queries, keys, values.

*

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* Multi-Headed Attention :→ enables transformers to captures diverse relationships in the data by learning multiple attention patterns simultaneousely, where each attention head computes self-attention independently, focusing on different parts of the input data.

* encoder :→ The purpose of encoder is to produce a contexual embeddings for each word in a given sentence.

* decoder :→ The purpose of decoder to produce an output sequence which can be a word or a sequence.

$\boxed{5}$

* Morphology :→ morphology is the study of the internal structure of words and how words are formed.
In NLP morphology helps computers understand Root words, prefixes, suffixes, word variations.
ex :→ Unhappiness un→ not, happy → root word, ness → state
* Morphemes :→ the smallest meaningfull unit in a Language.
cats → morphemes('cat' + 's')
(1) Free (alone) (2) Bound (not stand alone)
* add :→ attaching prefixes / suffixes to a root word.
happy → unhappy
* Delete operation :→ removing characters or morphemes.
making → make
Delete - ing.
* add/delete tables for word transformation :→